



Eco-Friendly Advances in Irrigation Practices

Ashamma¹, Srinivasa, V² and Nayana, K. R³

¹PG Scholar, Department of Vegetable Science, College of Horticulture, Mudigere

²Professor and Head, Department of Vegetable Science, College of Horticulture, Mudigere

³Assistant Professor, Department of Vegetable Science, College of Horticulture, Mudigere

Keladi Shivappa Nayaka University of Agricultural and Horticultural Sciences Iruvakk, Shivamogga

^{1*}Correspondence email: ashareddyv2002@gmail.com

Abstract

Rising food insecurity and hunger, coupled with the world’s rapidly growing population, demand a significant intensification of agricultural production. This need is particularly pressing in rural regions of India, especially those located in semi-arid and arid zones. In such areas, irrigation is vital not only to enhance crop reliability but also to ensure any agricultural output at all. With climate change driving more frequent and severe weather extremes-including prolonged dry spells, droughts and heat waves-the importance of irrigated agriculture in India is increasing even further. Beyond supporting food security, irrigated farming plays a crucial role in strengthening food value chains and fostering agribusiness opportunities in rural communities by enabling more stable and predictable production. However, sustainable management of limited water resources remains essential. Ensuring the economic viability of irrigation systems, setting fair water pricing and promoting efficient water distribution all call for strong planning, sound policy and capable implementation.

Keywords: Climate smart agriculture, Micro irrigation, Precision irrigation, Smart irrigation systems and Water use efficiency

Introduction

Water is the most vital resource for agriculture, yet it is becoming increasingly scarce due to climate change, population growth and over exploitation. In many parts of the world-especially in arid and semi-arid regions-farmers face the twin challenges of water scarcity and declining productivity. To overcome these challenges, modern innovations in irrigation are transforming how we use water in farming, making agriculture smarter, more efficient and more sustainable.

The Need for Innovation

Traditional irrigation methods like surface flooding and furrow irrigation often lead to water wastage through evaporation and runoff. With agriculture consuming nearly 70 per cent of the world’s freshwater resources, the need to shift towards efficient irrigation systems has never been more urgent. Innovative technologies now help farmers deliver the right amount of water at the right time and right place - saving both water and energy.

Drip Irrigation: Precision at the Roots

One of the most revolutionary methods is drip irrigation, which delivers water drop by drop directly to the root zone. This system ensures that every drop count. It not only saves up to 70% of water but also improves yields and reduces weed growth. In India, countries like Israel and the USA, drip irrigation has boosted productivity in crops such as sugarcane, grapes, vegetables and cotton.



Sprinkler Systems: Simulated Rain for Uniform Growth

Sprinkler irrigation sprays water in a way that mimics natural rainfall. It is especially useful for undulating land and sandy soils. Modern low-pressure sprinklers and centre pivot systems are designed to reduce energy use and ensure even water distribution. This system has gained popularity in countries like the United States and Brazil, where large-scale farming requires automation and efficiency.



Subsurface Drip Irrigation (SDI): Hidden Efficiency

An advanced form of drip irrigation, Subsurface Drip Irrigation (SDI) delivers water directly below the soil surface. It minimizes evaporation and salt build-up on the surface while maintaining optimal soil moisture. SDI is particularly effective for high-value crops in arid zones such as Rajasthan and Gujarat, where every drop of water is precious.



Smart and Automated Irrigation Systems

With the integration of Internet of Things (IoT), sensors and Artificial Intelligence (AI), irrigation has entered the digital age. Smart irrigation systems use soil moisture sensors, weather data and satellite monitoring to automatically control water delivery. Farmers can monitor and manage irrigation through smartphones or computers. For example, companies like Jain Irrigation and Netafim have developed smart controllers that ensure water is applied only when and where needed.



Solar-Powered Irrigation: Harnessing the Sun

Energy costs are a major challenge in irrigation. Solar-powered irrigation systems use solar panels to operate water pumps, making them eco-friendly and cost-effective. These systems are a boon for rural farmers who lack access to grid electricity. India’s PM-KUSUM scheme is a prime example, promoting solar pumps that empower farmers while reducing carbon emissions.



Use of Drones and Remote Sensing

Modern farmers are also adopting drones and remote sensing technologies to monitor soil moisture and crop health. This data-driven approach allows precise irrigation planning, reducing both water and fertilizer usage.



Recycled and Treated Wastewater

In regions facing extreme water shortages, treated wastewater is being reused for irrigation. Advanced filtration and treatment technologies ensure water quality and safety for crops. This method not only conserves freshwater but also helps in nutrient recycling.

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Conclusion

Innovations in irrigation represent the backbone of climate-smart agriculture. By adopting modern, efficient and sustainable methods, we can enhance food production while conserving our most precious resource-water. From smart sensors to solar pumps, these technologies empower farmers to achieve higher productivity with lower environmental impact.

In the years ahead, as climate challenges intensify, irrigation innovation will not just be a choice-it will be a necessity for ensuring global food security and sustainable growth.

References:

Food and Agriculture Organization (FAO),2017, The future of food and agriculture: Trends and challenges. *FAO*.

Food and Agriculture Organization (FAO) ,2021, Irrigation and water resources management. *FAO Water Reports*.

GOYAL, R. K. AND YADAV, R. K., 2014, Micro-irrigation systems and their role in enhancing water productivity. *Indian Journal of Agricultural Sciences*., 84(3): 321–328.

Jain Irrigation Systems Ltd. ,2020, Smart irrigation technologies and applications. Jain Irrigation Publications.

KUMAR, M. AND SINGH, R., 2020, Impact of drip irrigation on water use efficiency and crop productivity in arid regions. *Agricultural Water Management*, 239:106266.

Ministry of New and Renewable Energy (MNRE), 2022, *PM-KUSUM scheme guidelines*. Government of India.

Netafim.,2018, *Precision irrigation: Global innovations and impacts*. Netafim Knowledge Centre.

RAVISHANKAR, G. AND SOMASUNDARAM, J., 2018, Smart irrigation using IoT and sensors for sustainable agriculture. *Journal of Agricultural Informatic.s*., 9(2): 1–10.

SINGH, A. K., KUMAR, P. AND SHARMA, M., 2019, Subsurface drip irrigation: Advances, challenges and prospects. *Irrigation Science.*, 37(4), 469-483.

World Bank, 2020, *Water in agriculture: Challenges and opportunities for sustainable irrigation*. World Bank Group.

World Resources Institute (WRI) ,2019,*Wastewater reuse for agriculture: Technologies and policies for water-scarce regions*. WRI Research Paper.

